

02477 – Bayesian Machine Learning: Lecture 2

Michael Riis Andersen

Technical University of Denmark,
DTU Compute, Department of Applied Math and Computer Science

Outline

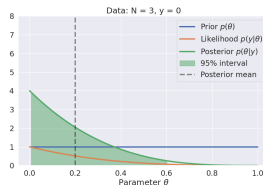
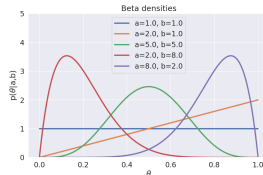
- 1 Quick re-cap of last week
- 2 Probabilistic machine learning
- 3 The plug-in approximation
- 4 Grid approximations for non-conjugate models
- 5 Introduction to exercise: towards logistic regression

Quick re-cap of last week

Quick re-cap of Beta-binomial model and what's next

- *Bayes' rule* provides a systematic way to combine data with prior knowledge

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$



Quick re-cap of Beta-binomial model and what's next

- *Bayes' rule* provides a systematic way to combine data with prior knowledge

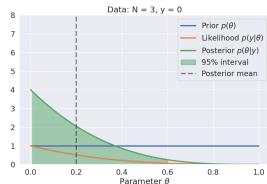
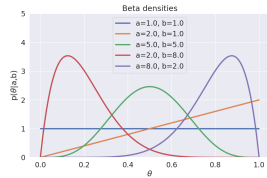
$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- *The beta-binomial model* is a *conjugate* model

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{N}{y} \theta^y (1 - \theta)^{N-y} \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0) \quad (\text{Posterior})$$



Quick re-cap of Beta-binomial model and what's next

- *Bayes' rule* provides a systematic way to combine data with prior knowledge

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- *The beta-binomial model* is a *conjugate* model

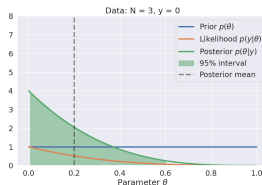
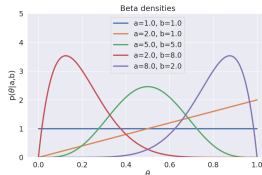
$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{N}{y} \theta^y (1 - \theta)^{N-y} \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0) \quad (\text{Posterior})$$

- Estimate θ using the *mean* of the *posterior distribution*

$$\theta_{\text{Bayes}} = \mathbb{E}[\theta|y] \equiv \int \theta p(\theta|y) d\theta$$



Quick re-cap of Beta-binomial model and what's next

- *Bayes' rule* provides a systematic way to combine data with prior knowledge

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- *The beta-binomial model* is a *conjugate* model

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{N}{y} \theta^y (1 - \theta)^{N-y} \quad (\text{Likelihood})$$

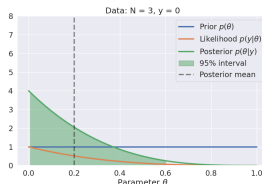
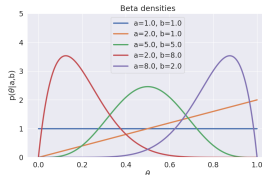
$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0) \quad (\text{Posterior})$$

- Estimate θ using the *mean* of the *posterior distribution*

$$\theta_{\text{Bayes}} = \mathbb{E}[\theta|y] \equiv \int \theta p(\theta|y) d\theta$$

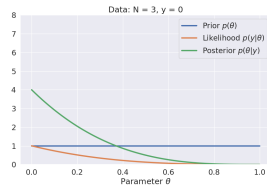
- .. and use *credibility intervals* of the posterior to quantify the uncertainty

$$P(\theta \in [0.01, 0.60] | y) = 0.95$$



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.



What about making predictions?

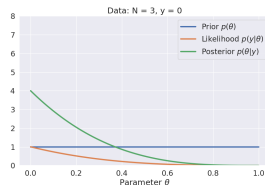
Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

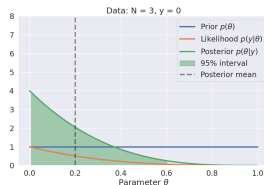
$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

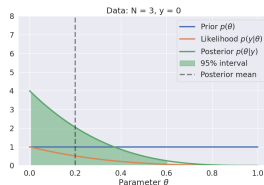
$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$

- **Goal:** *predict* number of clicks y^* in the next $N^* = 50$ views?

$$p(y^*|N^*, \theta) = \text{Bin}(y^*|N^*, \theta)$$



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

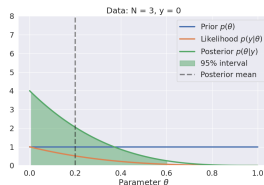
$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$

- **Goal:** *predict* number of clicks y^* in the next $N^* = 50$ views?

$$p(y^*|N^*, \theta) = \text{Bin}(y^*|N^*, \theta)$$

- Recall: mean of a Binomial random variable with prob. θ :

$$\mathbb{E}[y^*] = N^* \theta = 50\theta$$



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$

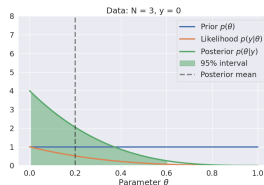
- **Goal:** *predict* number of clicks y^* in the next $N^* = 50$ views?

$$p(y^*|N^*, \theta) = \text{Bin}(y^*|N^*, \theta)$$

- Recall: mean of a Binomial random variable with prob. θ :

$$\mathbb{E}[y^*] = N^* \theta = 50\theta$$

Which value of θ to use?



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$

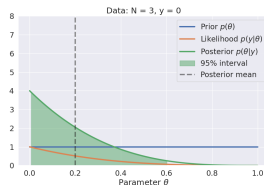
- **Goal:** *predict* number of clicks y^* in the next $N^* = 50$ views?

$$p(y^*|N^*, \theta) = \text{Bin}(y^*|N^*, \theta)$$

- Recall: mean of a Binomial random variable with prob. θ :

$$\mathbb{E}[y^*] = N^* \theta = 50\theta$$

Which value of θ to use? How do we use the *posterior knowledge* to make predictions?



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$

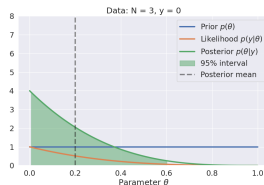
- **Goal:** *predict* number of clicks y^* in the next $N^* = 50$ views?

$$p(y^*|N^*, \theta) = \text{Bin}(y^*|N^*, \theta)$$

- Recall: mean of a Binomial random variable with prob. θ :

$$\mathbb{E}[y^*] = N^* \theta = 50\theta$$

Which value of θ to use? How do we use the *posterior knowledge* to make predictions?
How do we take the *uncertainty* into account?



Probabilistic machine learning

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)
- The *joint distribution* of all random variables can be expressed via the *product rule*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)
- The *joint distribution* of all random variables can be expressed via the *product rule*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- The *posterior distribution* can be obtained by *conditioning* on y

$$p(\theta|y) = \frac{p(y, \theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{p(y)}$$

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)
- The *joint distribution* of all random variables can be expressed via the *product rule*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- The *posterior distribution* can be obtained by *conditioning* on y

$$p(\theta|y) = \frac{p(y, \theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- The *evidence* $p(y)$ can be obtained from the joint distribution via the *sum rule*

$$p(y) = \int p(y, \theta)d\theta =$$

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)
- The *joint distribution* of all random variables can be expressed via the *product rule*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- The *posterior distribution* can be obtained by *conditioning* on y

$$p(\theta|y) = \frac{p(y, \theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- The *evidence* $p(y)$ can be obtained from the joint distribution via the *sum rule*

$$p(y) = \int p(y, \theta)d\theta = \int p(y|\theta)p(\theta)d\theta$$

- Hence, in theory, we can derive *all quantities of interest* from the *joint distribution*

Probabilistic machine learning II

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

Probabilistic machine learning II

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- What if we have more than one observed variable, e.g. y_1 and y_2 ?

Probabilistic machine learning II

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- What if we have more than one observed variable, e.g. y_1 and y_2 ?
- For a *broad class of models* the likelihood can be further decomposed using *conditional independence*:

$$p(y_1, y_2|\theta) = p(y_1|\theta)p(y_2|\theta)$$

Probabilistic machine learning II

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b}) d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- What if we have more than one observed variable, e.g. y_1 and y_2 ?
- For a *broad class of models* the likelihood can be further decomposed using *conditional independence*:

$$p(y_1, y_2|\theta) = p(y_1|\theta)p(y_2|\theta)$$

- ... or more generally for $\mathbf{y} = [y_1 \quad y_2 \quad \dots \quad y_N]$

$$p(\mathbf{y}|\theta) = p(y_1|\theta)p(y_2|\theta) \dots p(y_N|\theta) = \prod_{n=1}^N p(y_n|\theta)$$

Probabilistic machine learning II

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b}) d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- What if we have more than one observed variable, e.g. y_1 and y_2 ?
- For a *broad class of models* the likelihood can be further decomposed using *conditional independence*:

$$p(y_1, y_2|\theta) = p(y_1|\theta)p(y_2|\theta)$$

- ... or more generally for $\mathbf{y} = [y_1 \quad y_2 \quad \dots \quad y_N]$

$$p(\mathbf{y}|\theta) = p(y_1|\theta)p(y_2|\theta) \dots p(y_N|\theta) = \prod_{n=1}^N p(y_n|\theta)$$

- The *joint distribution* becomes

$$p(\theta, \mathbf{y}) = p(\mathbf{y}|\theta)p(\theta) = \prod_{n=1}^N p(y_n|\theta)p(\theta)$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta)$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta)$$

- The assumption of *conditional independence* implies

$$p(y, y^*|\theta) = p(y|\theta)p(y^*|\theta) =$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta)$$

- The assumption of *conditional independence* implies

$$p(y, y^*|\theta) = p(y|\theta)p(y^*|\theta) = \text{Bin}(y|N, \theta)\text{Bin}(y^*|N^*, \theta)$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta)$$

- The assumption of *conditional independence* implies

$$p(y, y^*|\theta) = p(y|\theta)p(y^*|\theta) = \text{Bin}(y|N, \theta)\text{Bin}(y^*|N^*, \theta)$$

- Completing the model by *imposing a prior* for θ

$$p(\theta) = \text{Beta}(\theta|a_0, b_0)$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta)$$

- The assumption of *conditional independence* implies

$$p(y, y^*|\theta) = p(y|\theta)p(y^*|\theta) = \text{Bin}(y|N, \theta)\text{Bin}(y^*|N^*, \theta)$$

- Completing the model by *imposing a prior* for θ

$$p(\theta) = \text{Beta}(\theta|a_0, b_0)$$

- **Goal:** compute *predictive distribution* of y^* given we have observed $y = 12$, i.e. $p(y^*|y = 12)$.

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) =$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} =$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

- Step 3: *Marginalize* out parameter θ using the *sum rule* to get the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*, \theta|y)d\theta =$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

- Step 3: *Marginalize* out parameter θ using the *sum rule* to get the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*, \theta|y)d\theta = \int \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}d\theta =$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

- Step 3: *Marginalize* out parameter θ using the *sum rule* to get the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*, \theta|y)d\theta = \int \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}d\theta = \int p(y^*|\theta)p(\theta|y)d\theta$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

- Step 3: *Marginalize* out parameter θ using the *sum rule* to get the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*, \theta|y)d\theta = \int \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}d\theta = \int p(y^*|\theta)p(\theta|y)d\theta = \mathbb{E}_{p(\theta|y)} [p(y^*|\theta)]$$

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

- Step 3: *Marginalize* out parameter θ using the *sum rule* to get the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*, \theta|y)d\theta = \int \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}d\theta = \int p(y^*|\theta)p(\theta|y)d\theta = \mathbb{E}_{p(\theta|y)} [p(y^*|\theta)]$$

- Key take-away:** To reason about y^* *given* y , we need to *average the likelihood* for y^* wrt. to the *posterior distribution* $p(\theta|y)$.

Quiz time

Take the quiz called
Lecture 2: Prior, likelihood, posterior, posterior predictive
to test your understanding.

Website example

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

Website example

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- We already defined the model

$$p(y|\theta) = \text{Bin}(y|N, \theta) \quad (\textit{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta) \quad (\textit{Predictive likelihood})$$

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\textit{Prior})$$

Website example

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- We already defined the model

$$p(y|\theta) = \text{Bin}(y|N, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

- We know how to compute the *posterior distribution*

$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0)$$

Website example

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- We already defined the model

$$p(y|\theta) = \text{Bin}(y|N, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

- We know how to compute the *posterior distribution*

$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0)$$

- Next, we want to compute the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta =$$

Website example

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

- We already defined the model

$$p(y|\theta) = \text{Bin}(y|N, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

- We know how to compute the *posterior distribution*

$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0)$$

- Next, we want to compute the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta = \int \text{Bin}(y^*|N^*, \theta)\text{Beta}(\theta|y + a_0, N - y + b_0)d\theta$$

- **Intuition:** Instead of plugging in a single value for the parameter estimate, we plug in all possible values for θ and weight the result according to $p(\theta|y)$

Website example

- Compute *posterior predictive distribution*

$$p(y^* = k|y) = \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta$$

Website example

- Compute *posterior predictive distribution*

$$\begin{aligned} p(y^* = k|y) &= \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta \\ &= \int \binom{N^*}{k} \theta^k (1 - \theta)^{N^* - k} \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \end{aligned}$$

Website example

- Compute *posterior predictive distribution*

$$\begin{aligned} p(y^* = k|y) &= \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta \\ &= \int \binom{N^*}{k} \theta^k (1 - \theta)^{N^* - k} \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \\ &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^k (1 - \theta)^{N^* - k} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \quad (\text{Linearity}) \end{aligned}$$

Website example

- Compute *posterior predictive distribution*

$$\begin{aligned} p(y^* = k|y) &= \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta \\ &= \int \binom{N^*}{k} \theta^k (1 - \theta)^{N^* - k} \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \\ &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^k (1 - \theta)^{N^* - k} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta && \text{(Linearity)} \\ &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta && \text{(Simplify)} \end{aligned}$$

Website example

- Compute *posterior predictive distribution*

$$\begin{aligned}
 p(y^* = k|y) &= \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta \\
 &= \int \binom{N^*}{k} \theta^k (1 - \theta)^{N^* - k} \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \\
 &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^k (1 - \theta)^{N^* - k} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta && \text{(Linearity)} \\
 &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta && \text{(Simplify)} \\
 &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta
 \end{aligned}$$

Website example

- Compute *posterior predictive distribution*

$$\begin{aligned}
 p(y^* = k|y) &= \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta \\
 &= \int \binom{N^*}{k} \theta^k (1 - \theta)^{N^* - k} \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \\
 &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^k (1 - \theta)^{N^* - k} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta && \text{(Linearity)} \\
 &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta && \text{(Simplify)} \\
 &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta
 \end{aligned}$$

- We recognize the terms in green as the *functional form* of a Beta density, and hence, we know how to compute the integral

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)}$$

for $\alpha = y + a_0$ and $\beta = N - y + b_0$.

Website example: putting everything together

Example continued: Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \text{Bin}(y|123, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|50, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|13, 112) \quad (\text{Posterior})$$

Website example: putting everything together

Example continued: Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \text{Bin}(y|123, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|50, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|13, 112) \quad (\text{Posterior})$$

- Distribution of clicks y^* based on views $N^* = 50$ views

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)} =$$

Website example: putting everything together

Example continued: Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \text{Bin}(y|123, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|50, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|13, 112) \quad (\text{Posterior})$$

- Distribution of clicks y^* based on views $N^* = 50$ views

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)} = \binom{50}{k} \frac{B(13 + k, 162 - k)}{B(13, 112)}$$

Website example: putting everything together

Example continued: Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

$$p(\theta) = \text{Beta}(\theta|1, 1)$$

(*Prior*)

$$p(y|\theta) = \text{Bin}(y|123, \theta)$$

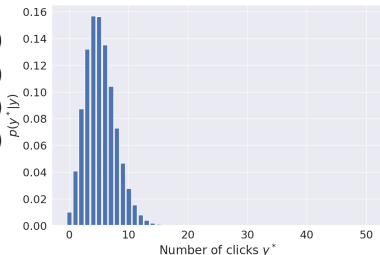
(*Likelihood*)

$$p(y^*|\theta) = \text{Bin}(y^*|50, \theta)$$

(*Predictive likelihood*)

$$p(\theta|y) = \text{Beta}(\theta|13, 112)$$

(*Posterior*)



- Distribution of clicks y^* based on views $N^* = 50$ views

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)} = \binom{50}{k} \frac{B(13 + k, 162 - k)}{B(13, 112)}$$

Website example: putting everything together

Example continued: Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

$$p(\theta) = \text{Beta}(\theta|1, 1)$$

(*Prior*)

$$p(y|\theta) = \text{Bin}(y|123, \theta)$$

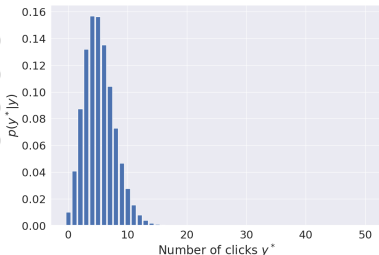
(*Likelihood*)

$$p(y^*|\theta) = \text{Bin}(y^*|50, \theta)$$

(*Predictive likelihood*)

$$p(\theta|y) = \text{Beta}(\theta|13, 112)$$

(*Posterior*)



- Distribution of clicks y^* based on views $N^* = 50$ views

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)} = \binom{50}{k} \frac{B(13 + k, 162 - k)}{B(13, 112)}$$

- The expected number of clicks given the data is

$$\mathbb{E}_{p(y^*|y)}[y^*] = \sum_{k=0}^{50} k p(y^* = k|y) \approx 5.2$$

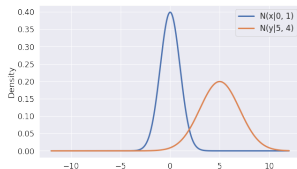
The plug-in approximation

A few prerequisites first

Univariate Gaussians

- The *normal distribution* (also known as the Gaussian) is distribution over $x \in \mathbb{R}$ with density

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



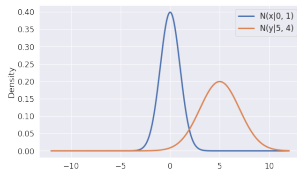
A few prerequisites first

Univariate Gaussians

- The *normal distribution* (also known as the Gaussian) is distribution over $x \in \mathbb{R}$ with density

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Two parameters: $\mu \equiv \mathbb{E}[x]$ and $\sigma^2 \equiv \mathbb{V}[x]$



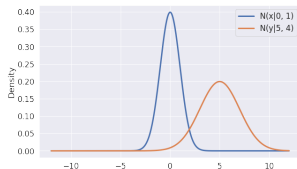
A few prerequisites first

Univariate Gaussians

- The *normal distribution* (also known as the Gaussian) is distribution over $x \in \mathbb{R}$ with density

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Two parameters: $\mu \equiv \mathbb{E}[x]$ and $\sigma^2 \equiv \mathbb{V}[x]$
- Widely popular due to Central limit theorems, maximum entropy principle, relation to least squares minimization, nice mathematical properties



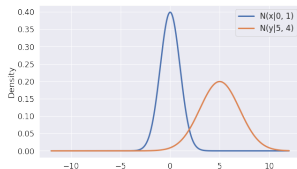
A few prerequisites first

Univariate Gaussians

- The *normal distribution* (also known as the Gaussian) is distribution over $x \in \mathbb{R}$ with density

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Two parameters: $\mu \equiv \mathbb{E}[x]$ and $\sigma^2 \equiv \mathbb{V}[x]$
- Widely popular due to Central limit theorems, maximum entropy principle, relation to least squares minimization, nice mathematical properties
- We will talk more about Gaussians later in this course



A few prerequisites first

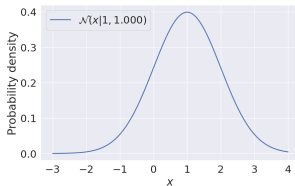
Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:

A few prerequisites first

Dirac's delta function

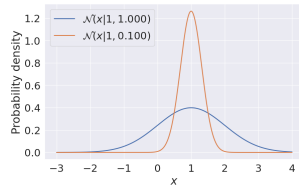
- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:



A few prerequisites first

Dirac's delta function

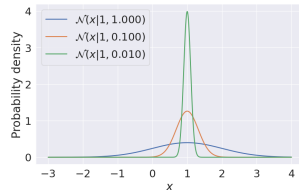
- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:



A few prerequisites first

Dirac's delta function

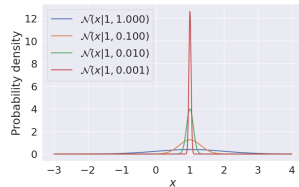
- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:



A few prerequisites first

Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:

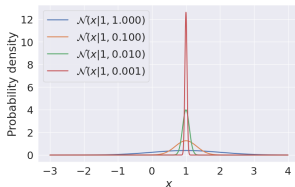


A few prerequisites first

Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:
- We say that x follows a *Dirac's delta distribution* centered at μ

$$p(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x|\mu, \sigma^2) = \delta(x - \mu)$$



A few prerequisites first

Dirac's delta function

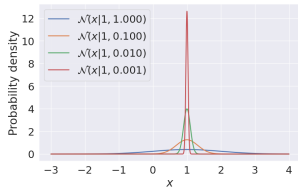
- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:
- We say that x follows a *Dirac's delta distribution* centered at μ

$$p(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x|\mu, \sigma^2) = \delta(x - \mu)$$

- Important properties

$$\delta(x - \mu) = \begin{cases} \infty & \text{if } x = \mu \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

(3)



A few prerequisites first

Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:
- We say that x follows a *Dirac's delta distribution* centered at μ

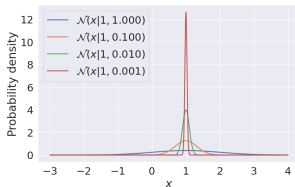
$$p(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x|\mu, \sigma^2) = \delta(x - \mu)$$

- Important properties

$$\delta(x - \mu) = \begin{cases} \infty & \text{if } x = \mu \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$\int \delta(x - \mu) dx = 1 \quad (2)$$

$$(3)$$



A few prerequisites first

Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:
- We say that x follows a *Dirac's delta distribution* centered at μ

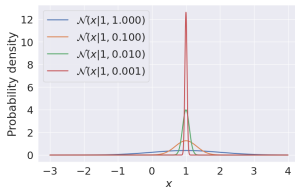
$$p(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x|\mu, \sigma^2) = \delta(x - \mu)$$

- Important properties

$$\delta(x - \mu) = \begin{cases} \infty & \text{if } x = \mu \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$\int \delta(x - \mu) dx = 1 \quad (2)$$

$$\int f(x) \delta(x - \mu) dx = f(\mu) \quad (3)$$



A few prerequisites first

Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:
- We say that x follows a *Dirac's delta distribution* centered at μ

$$p(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x|\mu, \sigma^2) = \delta(x - \mu)$$

- Important properties

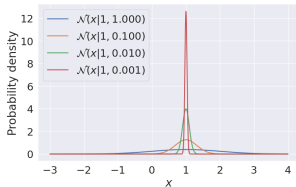
$$\delta(x - \mu) = \begin{cases} \infty & \text{if } x = \mu \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$\int \delta(x - \mu) dx = 1 \quad (2)$$

$$\int f(x) \delta(x - \mu) dx = f(\mu) \quad (3)$$

- Eq. (3) is called the *sifting property* and implies

$$\mathbb{E}_{\delta(x-\mu)} [f(x)] = f(\mu)$$



The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice

The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

- Using the *sifting property* of Dirac's delta implies

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta \approx \int p(y^*|\theta)\delta(\theta - \hat{\theta})d\theta = p(y^*|\hat{\theta})$$

The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

- Using the *sifting property* of Dirac's delta implies

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta \approx \int p(y^*|\theta)\delta(\theta - \hat{\theta})d\theta = p(y^*|\hat{\theta})$$

- Therefore, this is called a *plug-in approximation*.

The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

- Using the *sifting property* of Dirac's delta implies

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta \approx \int p(y^*|\theta)\delta(\theta - \hat{\theta})d\theta = p(y^*|\hat{\theta})$$

- Therefore, this is called a *plug-in approximation*.
- Very *easy to compute*, but *ignores uncertainty* for our estimate of θ , and hence, often *producing overconfident predictions*.

The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

- Using the *sifting property* of Dirac's delta implies

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta \approx \int p(y^*|\theta)\delta(\theta - \hat{\theta})d\theta = p(y^*|\hat{\theta})$$

- Therefore, this is called a *plug-in approximation*.
- Very *easy to compute*, but *ignores uncertainty* for our estimate of θ , and hence, often *producing overconfident predictions*.

The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

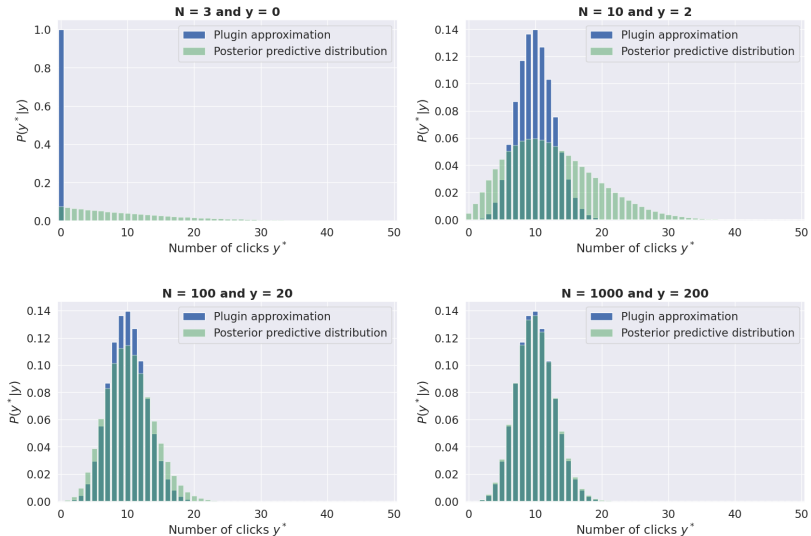
$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

- Using the *sifting property* of Dirac's delta implies

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta \approx \int p(y^*|\theta)\delta(\theta - \hat{\theta})d\theta = p(y^*|\hat{\theta})$$

- Therefore, this is called a *plug-in approximation*.
- Very *easy to compute*, but *ignores uncertainty* for our estimate of θ , and hence, often *producing overconfident predictions*.
- This is how we make predictions in deep learning...

The posterior predictive distribution and plugin approximations



Grid approximations for non-conjugate models

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta =$$

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

- To compute the posterior mean, variance etc. we need the evidence $p(y)$

$$p(y) = \int p(y|\theta)p(\theta)d\theta = \int \text{Bin}(y|N, \theta) \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta$$

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

- To compute the posterior mean, variance etc. we need the evidence $p(y)$

$$\begin{aligned} p(y) &= \int p(y|\theta)p(\theta)d\theta = \int \text{Bin}(y|N, \theta) \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \int \binom{N}{y} \theta^y (1-\theta)^{N-y} \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \end{aligned}$$

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

- To compute the posterior mean, variance etc. we need the evidence $p(y)$

$$\begin{aligned} p(y) &= \int p(y|\theta)p(\theta)d\theta = \int \text{Bin}(y|N, \theta) \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \int \binom{N}{y} \theta^y (1-\theta)^{N-y} \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \binom{N}{y} \frac{1}{Z} \int \theta^y (1-\theta)^{N-y} e^{\sin(\pi\theta^2)} d\theta = \end{aligned}$$

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

- To compute the posterior mean, variance etc. we need the evidence $p(y)$

$$\begin{aligned} p(y) &= \int p(y|\theta)p(\theta)d\theta = \int \text{Bin}(y|N, \theta) \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \int \binom{N}{y} \theta^y (1-\theta)^{N-y} \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \binom{N}{y} \frac{1}{Z} \int \theta^y (1-\theta)^{N-y} e^{\sin(\pi\theta^2)} d\theta = ? \end{aligned}$$

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

- To compute the posterior mean, variance etc. we need the evidence $p(y)$

$$\begin{aligned} p(y) &= \int p(y|\theta)p(\theta)d\theta = \int \text{Bin}(y|N, \theta) \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \int \binom{N}{y} \theta^y (1-\theta)^{N-y} \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \binom{N}{y} \frac{1}{Z} \int \theta^y (1-\theta)^{N-y} e^{\sin(\pi\theta^2)} d\theta = ? \end{aligned}$$

- Unfortunately, we *cannot* evaluate the evidence, i.e. $p(y)$ analytically *intractable* for most models of practical interest...

Approximate inference methods

- **In this course:** We will study several computational tools and approximation inference methods for dealing with such *intractable distributions*

1. Grid approximations
2. Laplace approximations
3. Variational inference
4. Markov Chain Monte Carlo methods

Approximate inference methods

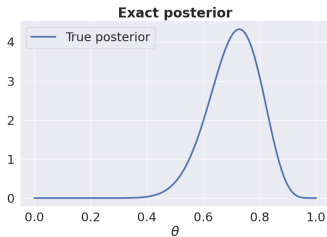
- **In this course:** We will study several computational tools and approximation inference methods for dealing with such *intractable distributions*
 1. Grid approximations
 2. Laplace approximations
 3. Variational inference
 4. Markov Chain Monte Carlo methods
- **Goal for these methods:** Compute *tractable approximation* $q(\theta)$ of true posterior $p(\theta|y)$ such that
 1. $q(\theta)$ resembles the true posterior, i.e. $p(\theta|y) \approx q(\theta)$
 2. $q(\theta)$ should be tractable s.t. we can compute posterior summaries, predictions etc.

Approximate inference methods

- **In this course:** We will study several computational tools and approximation inference methods for dealing with such *intractable distributions*
 1. Grid approximations
 2. Laplace approximations
 3. Variational inference
 4. Markov Chain Monte Carlo methods
- **Goal for these methods:** Compute *tractable approximation* $q(\theta)$ of true posterior $p(\theta|y)$ such that
 1. $q(\theta)$ resembles the true posterior, i.e. $p(\theta|y) \approx q(\theta)$
 2. $q(\theta)$ should be tractable s.t. we can compute posterior summaries, predictions etc.
- **This week:** We will focus on *grid approximations*, which are easy to understand and apply, and they will help build our intuition about marginalization.

The grid approximation

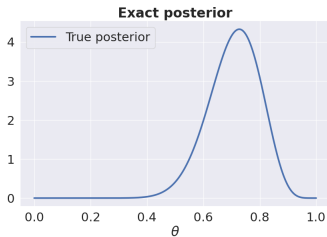
- Constructing the grid approximation for $p(\theta|y)$



The grid approximation

■ Constructing the grid approximation for $p(\theta|y)$

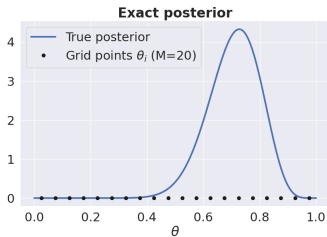
1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$



The grid approximation

■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$

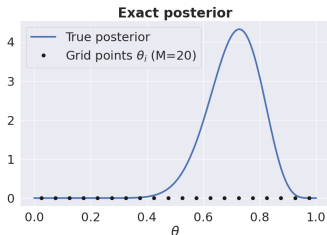


The grid approximation

■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$
2. We evaluate the exact posterior (up to a constant) at all the grid points, i.e.

$$\tilde{\pi}_i \propto p(\theta_i|y) \propto p(y|\theta_i)p(\theta_i)$$

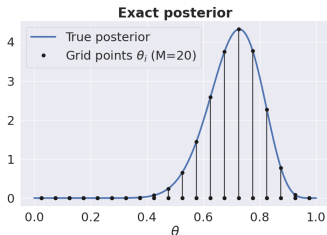


The grid approximation

■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$
2. We evaluate the exact posterior (up to a constant) at all the grid points, i.e.

$$\tilde{\pi}_i \propto p(\theta_i|y) \propto p(y|\theta_i)p(\theta_i)$$



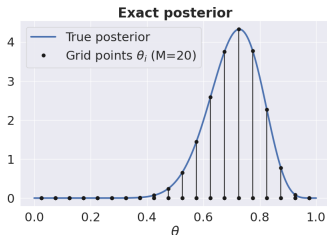
The grid approximation

■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$
2. We evaluate the exact posterior (up to a constant) at all the grid points, i.e.

$$\tilde{\pi}_i \propto p(\theta_i|y) \propto p(y|\theta_i)p(\theta_i)$$

3. Sum all values to get normalization constant $Z = \sum_{i=1}^M \tilde{\pi}_i$



The grid approximation

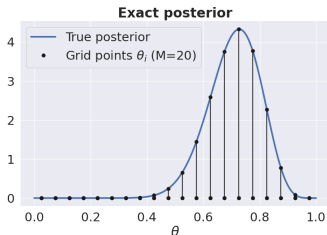
■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$
2. We evaluate the exact posterior (up to a constant) at all the grid points, i.e.

$$\tilde{\pi}_i \propto p(\theta_i|y) \propto p(y|\theta_i)p(\theta_i)$$

3. Sum all values to get normalization constant $Z = \sum_{i=1}^M \tilde{\pi}_i$
4. Compute normalized probabilities $\pi_i = \frac{1}{Z} \tilde{\pi}_i$ to get the grid approximation

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

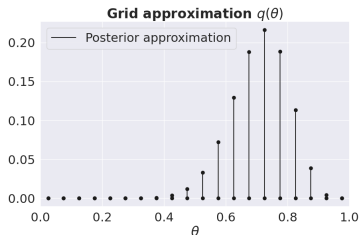
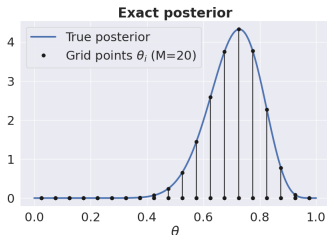
■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$
2. We evaluate the exact posterior (up to a constant) at all the grid points, i.e.

$$\tilde{\pi}_i \propto p(\theta_i|y) \propto p(y|\theta_i)p(\theta_i)$$

3. Sum all values to get normalization constant $Z = \sum_{i=1}^M \tilde{\pi}_i$
4. Compute normalized probabilities $\pi_i = \frac{1}{Z} \tilde{\pi}_i$ to get the grid approximation

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



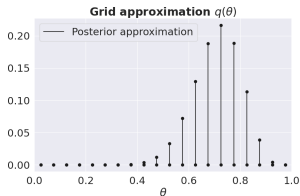
The grid approximation

Posterior summaries

- The grid approximation is a discrete distribution, so computing summaries is easy, e.g the posterior mean

$$\mathbb{E}_{p(\theta|y)} [\theta] \approx \mathbb{E}_{q(\theta)} [\theta] =$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



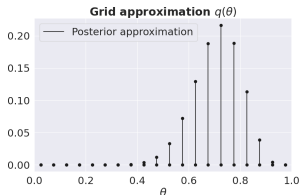
The grid approximation

Posterior summaries

- The grid approximation is a discrete distribution, so computing summaries is easy, e.g the posterior mean

$$\mathbb{E}_{p(\theta|y)} [\theta] \approx \mathbb{E}_{q(\theta)} [\theta] = \sum_{i=1}^M \theta_i \pi_i$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

Posterior summaries

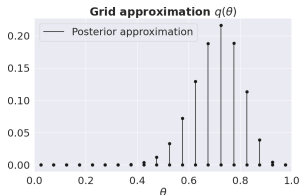
- The grid approximation is a discrete distribution, so computing summaries is easy, e.g the posterior mean

$$\mathbb{E}_{p(\theta|y)} [\theta] \approx \mathbb{E}_{q(\theta)} [\theta] = \sum_{i=1}^M \theta_i \pi_i$$

- ... and general expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

Posterior summaries

- The grid approximation is a discrete distribution, so computing summaries is easy, e.g the posterior mean

$$\mathbb{E}_{p(\theta|y)} [\theta] \approx \mathbb{E}_{q(\theta)} [\theta] = \sum_{i=1}^M \theta_i \pi_i$$

- ... and general expectations of $f(\theta)$

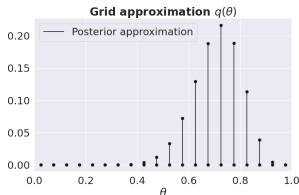
$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- Example: Computing post. probabilities for $\theta < 0.6$

$$p(\theta < 0.6|y) \approx q(\theta < 0.6)$$

$$= \sum_{i=1}^M \mathbb{I} [\theta_j < 0.6] \pi_j$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

Posterior summaries

- The grid approximation is a discrete distribution, so computing summaries is easy, e.g the posterior mean

$$\mathbb{E}_{p(\theta|y)} [\theta] \approx \mathbb{E}_{q(\theta)} [\theta] = \sum_{i=1}^M \theta_i \pi_i$$

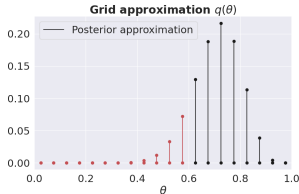
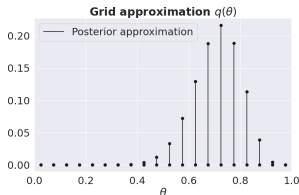
- ... and general expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- Example: Computing post. probabilities for $\theta < 0.6$

$$\begin{aligned} p(\theta < 0.6|y) &\approx q(\theta < 0.6) \\ &= \sum_{i=1}^M \mathbb{I}[\theta_i < 0.6] \pi_i \\ &= \sum_{i=1}^j \pi_i, \quad j = \max \{i | \theta_i < 0.6\} \end{aligned}$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



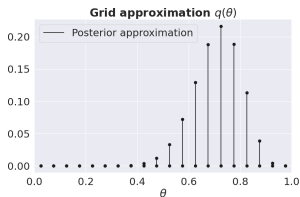
The grid approximation

The posterior predictive distribution

■ General expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

The posterior predictive distribution

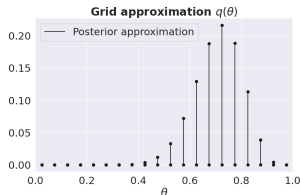
- General expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- The posterior predictive distribution $p(y^*|y)$

$$p(y^* = k|y) = \mathbb{E}_{p(\theta|y)} [\text{Bin}(y^* = k|N^*, \theta)]$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

The posterior predictive distribution

- General expectations of $f(\theta)$

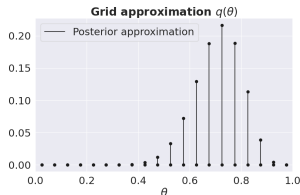
$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- The posterior predictive distribution $p(y^*|y)$

$$p(y^* = k|y) = \mathbb{E}_{p(\theta|y)} [\text{Bin}(y^* = k|N^*, \theta)]$$

- Hence, setting $f(\theta) = \text{Bin}(y^* = k|N^*, \theta_i)$ yields

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

The posterior predictive distribution

- General expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

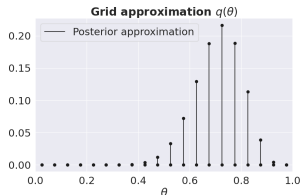
- The posterior predictive distribution $p(y^*|y)$

$$p(y^* = k|y) = \mathbb{E}_{p(\theta|y)} [\text{Bin}(y^* = k|N^*, \theta)]$$

- Hence, setting $f(\theta) = \text{Bin}(y^* = k|N^*, \theta_i)$ yields

$$p(y^* = k|y) \approx \sum_{i=1}^M \text{Bin}(y^* = k|N^*, \theta_i) \pi_i$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

The posterior predictive distribution

- General expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- The posterior predictive distribution $p(y^*|y)$

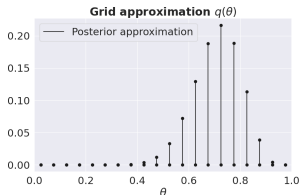
$$p(y^* = k|y) = \mathbb{E}_{p(\theta|y)} [\text{Bin}(y^* = k|N^*, \theta)]$$

- Hence, setting $f(\theta) = \text{Bin}(y^* = k|N^*, \theta_i)$ yields

$$p(y^* = k|y) \approx \sum_{i=1}^M \text{Bin}(y^* = k|N^*, \theta_i) \pi_i$$

- To make predictions we literally compute a weighted sum of all possible parameter values

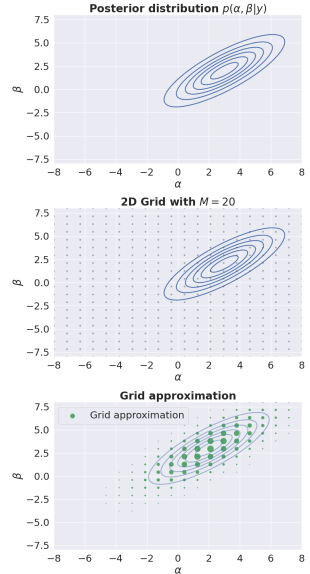
$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

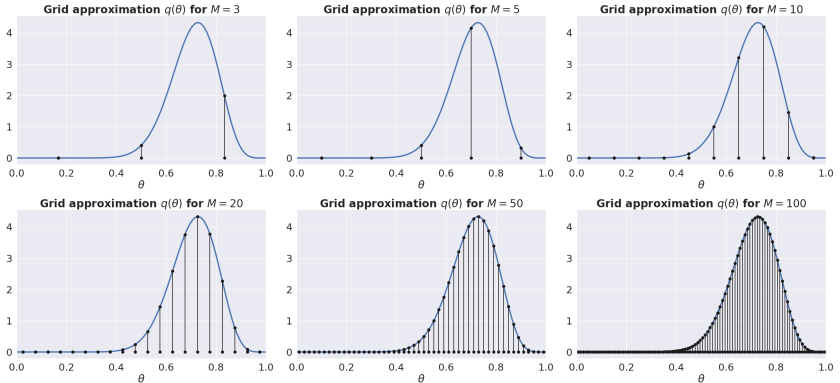
A few practical considerations

- Choosing the grid range
 - Range for $\theta \in [0, 1]$ is easy
 - For a different parameter $\alpha \in \mathbb{R}$, we need to choose an interval $[a, b]$ for the grid. Often identified visually.
- Scaling with model dimensionality
 - Suppose we use $M = 20$ points for each dimension
 - 1D: 20 evaluations
 - 2D: $20^2 = 400$ evaluations
 - 3D: $20^3 = 8000$ evaluations
 - 4D: $16000^4 = 160000$ evaluations
 - Grid approximations do not scale well beyond 3-4 dimensions
- Number of grid points M
 - M is balance between computational cost and accuracy
 - Grid approximation is zero when evaluated outside the grid points
 - Often diminishing returns as M increases (next slide)



The grid approximation

Summary



- **Pros:** Simple, easy and intuitive.
- **Cons:** Suffers from curse of dimensionality and does not scale beyond 3-4 dimensions

Introduction to exercise: towards logistic regression

Towards logistic regression

- So far we focussed on modelling *proportions*, i.e. $\theta \in [0, 1]$ given data about y successes in N *conditionally independent trials*
- The binomial likelihood is also often used in *dose-response* models, which is key for determining "safe" dosages for drugs, pollution, foods etc.

Towards logistic regression

- So far we focussed on modelling *proportions*, i.e. $\theta \in [0, 1]$ given data about y successes in N *conditionally independent trials*
- The binomial likelihood is also often used in *dose-response* models, which is key for determining "safe" dosages for drugs, pollution, foods etc.
- **Example:** A company wants to study side effects of their new drug

x (Dose in mg)	y (# side effects)	N (# patients)
80	0	69
160	4	832
320	13	835
480	20	459
640	12	324
800	6	103

Towards logistic regression

- So far we focussed on modelling *proportions*, i.e. $\theta \in [0, 1]$ given data about y successes in N *conditionally independent trials*
- The binomial likelihood is also often used in *dose-response* models, which is key for determining "safe" dosages for drugs, pollution, foods etc.
- **Example:** A company wants to study side effects of their new drug

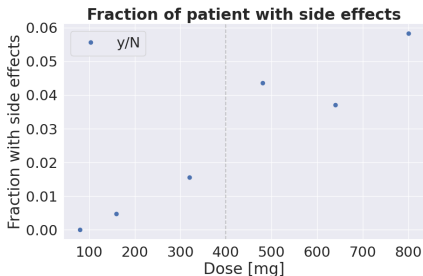
x (Dose in mg)	y (# side effects)	N (# patients)
80	0	69
160	4	832
320	13	835
480	20	459
640	12	324
800	6	103

- We could analyze the data for each dose independently using a beta-binomial model, but we would like to ...
 1. understand how dose affect the probability of side effects
 2. make predictions for new dosages x^*
 3. borrow "statistical strength" across dosages

Towards logistic regression

Example & motivation II

x (Dose in mg)	y (# side effects)	N (# patients)	y/N
80	0	69	0
160	4	832	0.005
320	13	835	0.016
480	20	459	0.044
640	12	324	0.037
800	6	103	0.058



- How accurate can we predict the probability of side effects for $x^* = 400\text{mg}$?

Towards logistic regression

Setting up the likelihood

- For each dose x_i , we assume

$$y_i|x_i \sim \text{Bin}(y_i|N_i, \theta_i), \quad \theta_i \in [0, 1]$$

Towards logistic regression

Setting up the likelihood

- For each dose x_i , we assume

$$y_i | x_i \sim \text{Bin}(y_i | N_i, \theta_i), \quad \theta_i \in [0, 1]$$

- We model the probability θ_i as function of the dose x_i , i.e.

$$\theta_i \equiv \theta(x_i) = \sigma(\alpha + \beta x),$$

where $\sigma(x) : \mathbb{R} \rightarrow [0, 1]$ is a sigmoid function and $\alpha, \beta \in \mathbb{R}$ are model parameters.

Towards logistic regression

Setting up the likelihood

- For each dose x_i , we assume

$$y_i | x_i \sim \text{Bin}(y_i | N_i, \theta_i), \quad \theta_i \in [0, 1]$$

- We model the probability θ_i as function of the dose x_i , i.e.

$$\theta_i \equiv \theta(x_i) = \sigma(\alpha + \beta x),$$

where $\sigma(x) : \mathbb{R} \rightarrow [0, 1]$ is a sigmoid function and $\alpha, \beta \in \mathbb{R}$ are model parameters.

- The likelihood of the i 'th observation (x_i, y_i)

$$p(y_i | x_i, \alpha, \beta) = \text{Bin}(y_i | N_i, \theta_i)$$

Towards logistic regression

Setting up the likelihood

- For each dose x_i , we assume

$$y_i|x_i \sim \text{Bin}(y_i|N_i, \theta_i), \quad \theta_i \in [0, 1]$$

- We model the probability θ_i as function of the dose x_i , i.e.

$$\theta_i \equiv \theta(x_i) = \sigma(\alpha + \beta x_i),$$

where $\sigma(x) : \mathbb{R} \rightarrow [0, 1]$ is a sigmoid function and $\alpha, \beta \in \mathbb{R}$ are model parameters.

- The likelihood of the i 'th observation (x_i, y_i)

$$p(y_i|x_i, \alpha, \beta) = \text{Bin}(y_i|N_i, \theta_i)$$

- Assuming conditional independence we can write the joint likelihood

$$p(\mathbf{y}|\mathbf{x}, \alpha, \beta) = \prod_{i=1}^M p(y_i|x_i, \alpha, \beta) = \prod_{i=1}^M \text{Bin}(y_i|N_i, \theta_i),$$

where $\mathbf{y} = [y_1, y_2, \dots, y_6]$ and similar for $\mathbf{x} = [x_1, x_2, \dots, x_6]$

Towards logistic regression

Setting up the likelihood

- For each dose x_i , we assume

$$y_i | x_i \sim \text{Bin}(y_i | N_i, \theta_i), \quad \theta_i \in [0, 1]$$

- We model the probability θ_i as function of the dose x_i , i.e.

$$\theta_i \equiv \theta(x_i) = \sigma(\alpha + \beta x_i),$$

where $\sigma(x) : \mathbb{R} \rightarrow [0, 1]$ is a sigmoid function and $\alpha, \beta \in \mathbb{R}$ are model parameters.

- The likelihood of the i 'th observation (x_i, y_i)

$$p(y_i | x_i, \alpha, \beta) = \text{Bin}(y_i | N_i, \theta_i)$$

- Assuming conditional independence we can write the joint likelihood

$$p(\mathbf{y} | \mathbf{x}, \alpha, \beta) = \prod_{i=1}^M p(y_i | x_i, \alpha, \beta) = \prod_{i=1}^M \text{Bin}(y_i | N_i, \theta_i),$$

where $\mathbf{y} = [y_1, y_2, \dots, y_6]$ and similar for $\mathbf{x} = [x_1, x_2, \dots, x_6]$

- The predictive likelihood for y^* is

$$p(y^* | x^*, \alpha, \beta) = \text{Bin}(y^* | N_i, \theta^*)$$

where $\theta^* \equiv \theta(x^*)$

Towards logistic regression

Setting up the prior

- The model parameters are α (intercept) and β (slope) of the generalized linear model

Towards logistic regression

Setting up the prior

- The model parameters are α (intercept) and β (slope) of the generalized linear model
- Prior information: we have no prior information about the sign of the parameters. Hence, we choose a zero-mean Gaussian distributions

$$p(\alpha, \beta) = \mathcal{N}(\alpha|0, \sigma_\alpha^2)\mathcal{N}(\beta|0, \sigma_\beta^2), \quad \sigma_\alpha^2, \sigma_\beta^2 > 0$$

Towards logistic regression

Setting up the prior

- The model parameters are α (intercept) and β (slope) of the generalized linear model
- Prior information: we have no prior information about the sign of the parameters. Hence, we choose a zero-mean Gaussian distributions

$$p(\alpha, \beta) = \mathcal{N}(\alpha|0, \sigma_\alpha^2)\mathcal{N}(\beta|0, \sigma_\beta^2), \quad \sigma_\alpha^2, \sigma_\beta^2 > 0$$

- We can now write the *joint distribution* of $\alpha, \beta, \mathbf{y}, y^*$ using the *product rule*

$$p(\mathbf{y}, y^*, \alpha, \beta | \mathbf{x}, x^*) = p(y^* | x^*, \alpha, \beta) p(\mathbf{y} | \mathbf{x}, \alpha, \beta) p(\alpha, \beta)$$

Towards logistic regression

Setting up the prior

- The model parameters are α (intercept) and β (slope) of the generalized linear model
- Prior information: we have no prior information about the sign of the parameters. Hence, we choose a zero-mean Gaussian distributions

$$p(\alpha, \beta) = \mathcal{N}(\alpha|0, \sigma_\alpha^2)\mathcal{N}(\beta|0, \sigma_\beta^2), \quad \sigma_\alpha^2, \sigma_\beta^2 > 0$$

- We can now write the *joint distribution* of $\alpha, \beta, \mathbf{y}, y^*$ using the *product rule*

$$p(\mathbf{y}, y^*, \alpha, \beta | \mathbf{x}, x^*) = p(y^* | x^*, \alpha, \beta) p(\mathbf{y} | \mathbf{x}, \alpha, \beta) p(\alpha, \beta)$$

- The *posterior predictive distribution* is by *conditioning on $\mathbf{y}$* and *marginalizing out the parameters* via the sum rule

$$p(y^* | \mathbf{y}, \mathbf{x}, x^*) = \iint p(y^*, \alpha, \beta | \mathbf{y}, \mathbf{x}, x^*) d\alpha d\beta =$$

Towards logistic regression

Setting up the prior

- The model parameters are α (intercept) and β (slope) of the generalized linear model
- Prior information: we have no prior information about the sign of the parameters. Hence, we choose a zero-mean Gaussian distributions

$$p(\alpha, \beta) = \mathcal{N}(\alpha|0, \sigma_\alpha^2) \mathcal{N}(\beta|0, \sigma_\beta^2), \quad \sigma_\alpha^2, \sigma_\beta^2 > 0$$

- We can now write the *joint distribution* of $\alpha, \beta, \mathbf{y}, \mathbf{y}^*$ using the *product rule*

$$p(\mathbf{y}, \mathbf{y}^*, \alpha, \beta | \mathbf{x}, \mathbf{x}^*) = p(\mathbf{y}^* | \mathbf{x}^*, \alpha, \beta) p(\mathbf{y} | \mathbf{x}, \alpha, \beta) p(\alpha, \beta)$$

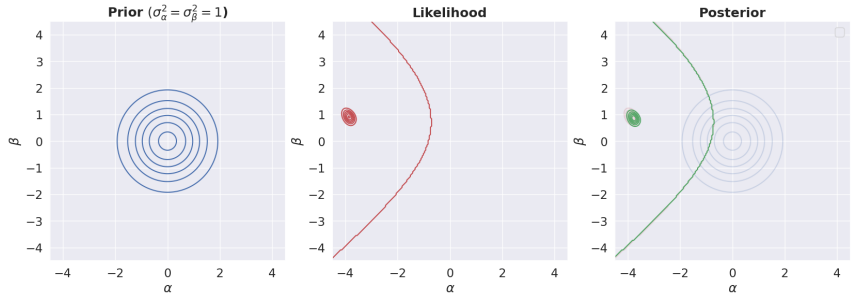
- The *posterior predictive distribution* is by *conditioning on $\mathbf{y}$* and *marginalizing out the parameters* via the sum rule

$$p(\mathbf{y}^* | \mathbf{y}, \mathbf{x}, \mathbf{x}^*) = \iint p(\mathbf{y}^*, \alpha, \beta | \mathbf{y}, \mathbf{x}, \mathbf{x}^*) d\alpha d\beta = \iint \underbrace{p(\mathbf{y}^* | \mathbf{x}^*, \alpha, \beta)}_{\text{likelihood for } \mathbf{y}^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta$$

- After obtaining the posterior of α, β , we can *propagate* the posterior uncertainty of the parameters to any quantity that depends on α, β , i.e. $\theta(x) = \sigma(\alpha + \beta x)$, the fraction of people with side effects $\mathbf{y}^*/N$ etc.

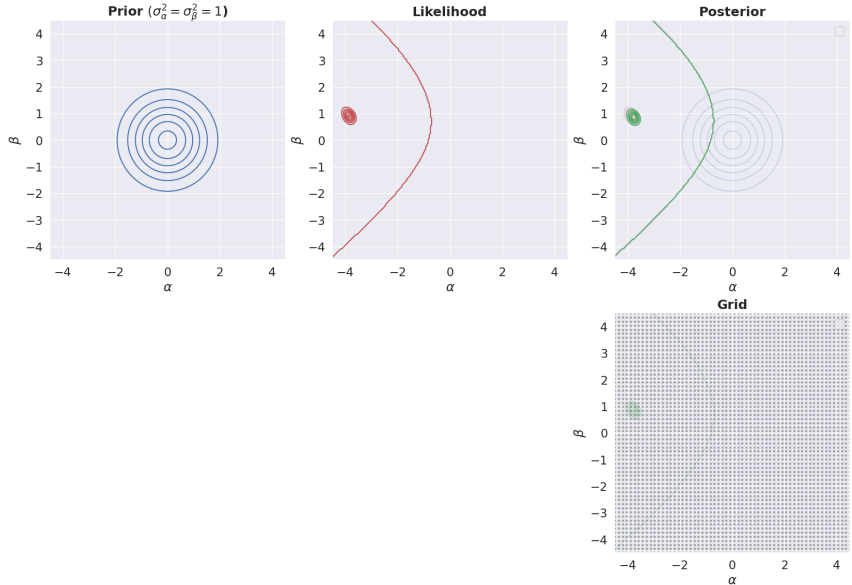
Towards logistic regression

Visualizing the distributions



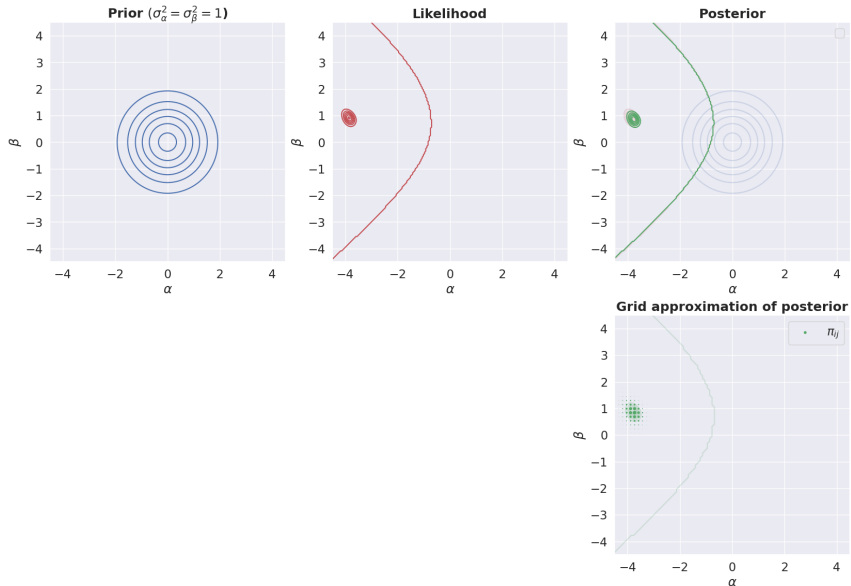
Towards logistic regression

Visualizing the distributions



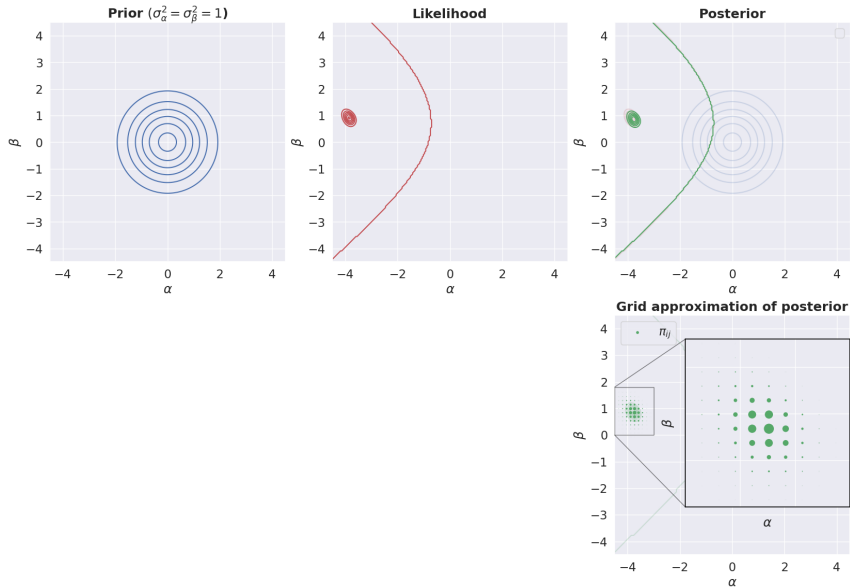
Towards logistic regression

Visualizing the distributions



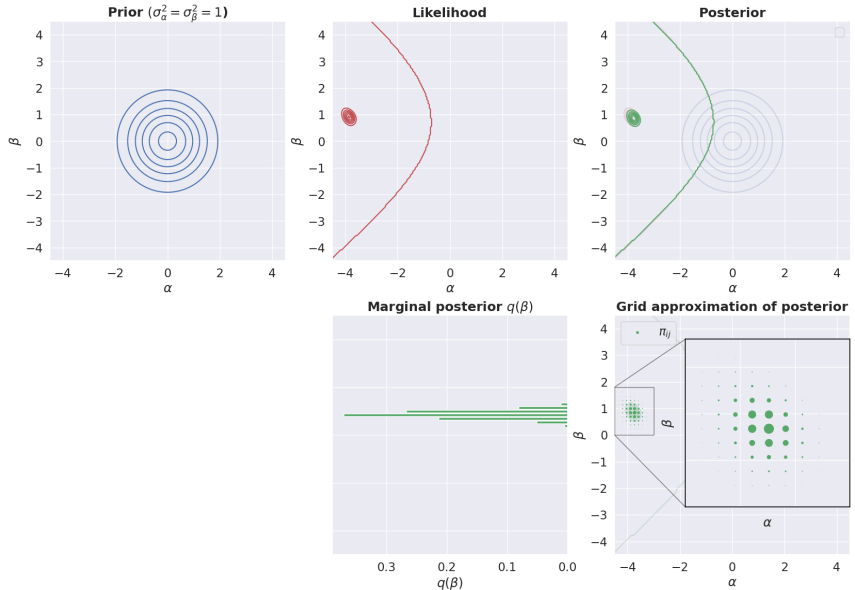
Towards logistic regression

Visualizing the distributions



Towards logistic regression

Visualizing the distributions



Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) = \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta \quad (\text{Sum rule})$$

Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned} p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\ &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \end{aligned}$$

Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned} p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\ &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\ &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \end{aligned}$$

Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned}
 p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \\
 &\approx \mathbb{E}_{q(\alpha, \beta)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Grid approx.)}
 \end{aligned}$$

Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

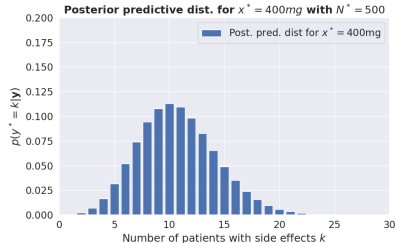
$$\begin{aligned}
 p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \\
 &\approx \mathbb{E}_{q(\alpha, \beta)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Grid approx.)} \\
 &= \sum_{i,j} \text{Bin}(y^* | N^*, \sigma(\alpha_i + \beta_j x^*)) \pi_{ij}
 \end{aligned}$$

Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned}
 p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \\
 &\approx \mathbb{E}_{q(\alpha, \beta)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Grid approx.)} \\
 &= \sum_{i,j} \text{Bin}(y^* | N^*, \sigma(\alpha_i + \beta_j x^*)) \pi_{ij}
 \end{aligned}$$

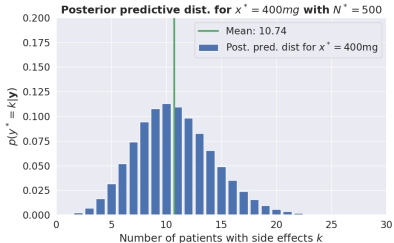


Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned}
 p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \\
 &\approx \mathbb{E}_{q(\alpha, \beta)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Grid approx.)} \\
 &= \sum_{i,j} \text{Bin}(y^* | N^*, \sigma(\alpha_i + \beta_j x^*)) \pi_{ij}
 \end{aligned}$$

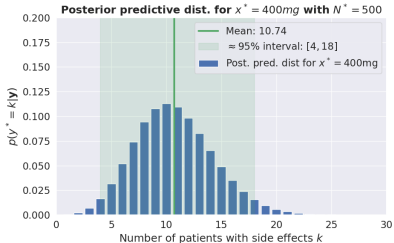


Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned}
 p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\
 &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \\
 &\approx \mathbb{E}_{q(\alpha, \beta)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Grid approx.)} \\
 &= \sum_{i,j} \text{Bin}(y^* | N^*, \sigma(\alpha_i + \beta_j x^*)) \pi_{ij}
 \end{aligned}$$



Intro to exercise

- On DTU Learn you will find an exercise for each week in notebook format
- We will spend all 4 hours from 13-17 working with the exercises
- In this exercise you will
 - Dive deeper into the Bayesian framework
 - Study and implement the probabilistic model for logistic regression for the Challenger Distaster dataset
 - Study and implement the grid approximations
 - Practice probabilistic reasoning
- Mix of pen&paper, programming and discussion questions
- Feel free to collaborate with your peers
- Ask for help!
 - Ask for help when stuck
 - Use teachers/TAs to check your understanding
 - Engage in discussion to practice
- Feedbacks persons: Meet at 16:45